Trace Estimation of PSD MPOs via Tensor Network Sketching

1 Introduction and Setup

Let $A \in \mathbb{R}^{d^N \times d^N}$ be a Positive Semi-Definite (PSD) Matrix Product Operator (MPO) defined on N sites, with physical dimension d and bond dimension D. As A is PSD, it admits the spectral decomposition:

$$A = \sum_{i=1}^{d^N} \lambda_i u_i u_i^T, \tag{1}$$

where $\lambda_i \geq 0$ are the eigenvalues and $u_i \in \mathbb{R}^{d^N}$ are the corresponding orthonormal eigenvectors.

We are interested in estimating the exact trace of A:

$$\mathrm{Tr}(A) = \sum_{i=1}^{d^N} \lambda_i. \tag{2}$$

We estimate this trace using a sketching matrix $S \in \mathbb{R}^{m \times d^N}$ formulated as a chain-structured tensor network. The sketched matrix is $SAS^T \in \mathbb{R}^{m \times m}$, and our estimator is $\mathrm{Tr}(SAS^T)$.

1.1 Proof of the (ϵ, δ)-Embedding for $\mathrm{Tr}(SAS^T)$

Theorem 1. *Given a chain-structured sketching matrix $S \in \mathbb{R}^{m \times d^N}$ that satisfies the (ϵ, δ')-embedding property from Theorem 3.2 of Ma and Solomonik (arXiv:2205.13163) for $\delta' = \frac{\delta}{d^N}$, the trace of the sketched MPO satisfies:*

$$\mathbb{P}\left(\left|\mathrm{Tr}(SAS^T) - \mathrm{Tr}(A)\right| \leq \epsilon\, \mathrm{Tr}(A)\right) \geq 1 - \delta. \tag{3}$$

Furthermore, the required sketching dimension scales as $m = \Theta\left(\frac{N^2 \log d + N \log(1/\delta)}{\epsilon^2}\right)$.

Proof. First, we expand the trace of the sketched matrix SAS^T using the spectral decomposition of A and the linearity of the trace operator:

$$\mathrm{Tr}(SAS^T) = \mathrm{Tr}\left(S\left(\sum_{i=1}^{d^N} \lambda_i u_i u_i^T\right) S^T\right) \tag{4}$$

$$= \sum_{i=1}^{d^N} \lambda_i\, \mathrm{Tr}\left(S u_i u_i^T S^T\right) \tag{5}$$

$$= \sum_{i=1}^{d^N} \lambda_i\, \mathrm{Tr}\left(u_i^T S^T S u_i\right) \tag{6}$$

$$= \sum_{i=1}^{d^N} \lambda_i \|S u_i\|_2^2. \tag{7}$$

By Theorem 3.2 in arXiv:2205.13163, the sketching matrix S provides an (ϵ, δ')-accurate embedding for any fixed vector. Therefore, for a fixed eigenvector u_i, we have:

$$\mathbb{P}\left(\left|\|Su_i\|_2^2 - \|u_i\|_2^2\right| \leq \epsilon \|u_i\|_2^2\right) \geq 1 - \delta'. \tag{8}$$

Since u_i is an orthonormal eigenvector, $\|u_i\|_2^2 = 1$, which simplifies the bound to:

$$\mathbb{P}\left(\left|\|Su_i\|_2^2 - 1\right| \leq \epsilon\right) \geq 1 - \delta'. \tag{9}$$

To ensure this bound holds simultaneously for all d^N eigenvectors of A, we apply a union bound. Let E_i be the failure event where $\left|\|Su_i\|_2^2 - 1\right| > \epsilon$. We know that $\mathbb{P}(E_i) \leq \delta'$. The probability that at least one such failure event occurs is bounded by:

$$\mathbb{P}\left(\bigcup_{i=1}^{d^N} E_i\right) \leq \sum_{i=1}^{d^N} \mathbb{P}(E_i) \leq d^N \delta'. \tag{10}$$

By choosing our target failure probability such that $d^N\delta' = \delta$ (meaning $\delta' = \frac{\delta}{d^N}$), the total probability of failure is bounded by δ. Thus, with probability at least $1-\delta$, the condition $\left|\|Su_i\|_2^2 - 1\right| \leq \epsilon$ holds for all $i \in \{1, \ldots, d^N\}$ simultaneously.

Conditioned on this success event, we bound the summation in Equation 7. Substituting the bounds $1-\epsilon \leq \|Su_i\|_2^2 \leq 1+\epsilon$ yields:

$$\sum_{i=1}^{d^N} \lambda_i(1-\epsilon) \leq \operatorname{Tr}(SAS^T) \leq \sum_{i=1}^{d^N} \lambda_i(1+\epsilon) \tag{11}$$

$$(1-\epsilon)\sum_{i=1}^{d^N} \lambda_i \leq \operatorname{Tr}(SAS^T) \leq (1+\epsilon)\sum_{i=1}^{d^N} \lambda_i \tag{12}$$

$$(1-\epsilon)\operatorname{Tr}(A) \leq \operatorname{Tr}(SAS^T) \leq (1+\epsilon)\operatorname{Tr}(A). \tag{13}$$

Subtracting $\operatorname{Tr}(A)$ gives:

$$\left|\operatorname{Tr}(SAS^T) - \operatorname{Tr}(A)\right| \leq \epsilon \operatorname{Tr}(A), \tag{14}$$

which establishes the ϵ-accuracy of the estimator.

Derivation of the Sketching Dimension m: From Theorem 3.2 of Ma and Solomonik, an (ϵ, δ')-accurate embedding using a chain-structured tensor network sketch S applied to an N-dimensional data tensor requires a sketch size m satisfying:

$$m = \Theta\left(\frac{N\log(1/\delta')}{\epsilon^2}\right). \tag{15}$$

Substituting the required per-vector failure probability $\delta' = \frac{\delta}{d^N}$ dictated by the union bound over the full eigenspace yields the final required sketching dimension:

$$m = \Theta\left(\frac{N\log\left(\frac{d^N}{\delta}\right)}{\epsilon^2}\right) \tag{16}$$

$$= \Theta\left(\frac{N\left(N\log d + \log(1/\delta)\right)}{\epsilon^2}\right) \tag{17}$$

$$= \Theta\left(\frac{N^2\log d + N\log(1/\delta)}{\epsilon^2}\right). \tag{18}$$

Equation 18 demonstrates that the sketching dimension m must scale quadratically with the number of tensor sites N and logarithmically with the physical dimension d to maintain the overall (ϵ, δ)-accuracy constraint. Because the union bound evaluates the entire d^N-dimensional Hilbert space directly, the specific bond dimension D of the original MPO A does not explicitly alter the required dimension m in this worst-case spectral bounding framework. □

1.2 Sample Complexity for Estimating $\operatorname{Tr}(A^k)$, $k \geq 2$

Throughout this subsection $n := d^N$ and $0 \neq A = U\Lambda U^T \succeq 0$, where $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, $\lambda_i \geq 0$, and $U = [u_1, \ldots, u_n]$ is orthogonal. Let $S_1, \ldots, S_k \in \mathbb{R}^{m\times n}$ be sketching matrices. Cycling the trace and conjugating by U, the estimator is

$$E_k \;=\; \operatorname{Tr}\big(S_1 A S_2^T S_2 A \cdots S_k A S_1^T\big) \;=\; \operatorname{Tr}\Big(\prod_{r=1}^{k} S_r^T S_r\, A\Big) \;=\; \operatorname{Tr}\Big(\prod_{r=1}^{k} M_r \Lambda\Big), \qquad M_r := U^T S_r^T S_r U. \tag{19}$$

Write $\Gamma_r := M_r - I$, so that $(\Gamma_r)_{ij} = \langle S_r u_i, S_r u_j\rangle - \delta_{ij}$, and define the *intrinsic dimension*

$$\rho \;:=\; \frac{\mathrm{Tr}(A)}{\|A\|_2} \;\in\; \big[1, \mathrm{rank}(A)\big]. \tag{20}$$

No probabilistic assumption beyond the following is used. Independence of $S_1, \dots, S_k$ is not required by any proof below; it enters only through the unbiasedness identity $\mathbb{E}[E_k] = \mathrm{Tr}(A^k)$ (valid when $\mathbb{E}[S_r^T S_r] = I$), which is not needed.

Definition 2 (Embedding property)**.** *$S \in \mathbb{R}^{m\times n}$ has the (ϵ, δ')-*embedding property *if for every fixed $x \in \mathbb{R}^n$,*

$$\mathbb{P}\Big(\big|\, \|Sx\|_2^2 - \|x\|_2^2 \,\big| > \epsilon \|x\|_2^2\Big) \;\le\; \delta'. \tag{21}$$

By Corollary 3.2 of Ma–Solomonik (arXiv:2205.13163), a Gaussian chain-structured (tensor train) embedding with a single output sketch dimension and all internal dimensions of size m has this property, with $m = \Theta\big(N \log(1/\delta')/\epsilon^2\big)$ sufficient.

Theorem 3. *Let $k \ge 2$, $\hat\epsilon \in (0, 1]$, $\delta \in (0, 1)$, and let each S_r have the (ϵ, δ')-embedding property with*

$$\delta' \;=\; \frac{\delta}{k\, d^{2N}}, \qquad \epsilon \;=\; \min\Big\{\frac{\hat\epsilon}{2k},\; \frac{1}{k\rho}\sqrt{\hat\epsilon/2}\Big\}. \tag{22}$$

Then

$$\mathbb{P}\Big(\big|E_k - \mathrm{Tr}(A^k)\big| \;\le\; \hat\epsilon\, \mathrm{Tr}(A^k)\Big) \;\ge\; 1 - \delta. \tag{23}$$

With the chain-structured sketch of Definition 2 it therefore suffices to take

$$m \;=\; \Theta\bigg(\max\Big\{\frac{4k^2}{\hat\epsilon^2},\; \frac{2k^2\rho^2}{\hat\epsilon}\Big\} \cdot N \log \frac{k\, d^{2N}}{\delta}\bigg) \;=\; O\bigg(\frac{k^2 \max\{1,\; \rho^2\hat\epsilon\}}{\hat\epsilon^2}\,\big(N^2 \log d + N \log(k/\delta)\big)\bigg). \tag{24}$$

The proof consists of two lemmas: a two-sided high-probability control of the entries of Γ_r, and a deterministic deviation inequality that is exact at the level of element-wise information.

Lemma 4 (Two-sided control of the sketched Gram matrices)**.** *If each S_r has the (ϵ, δ')-embedding property, then with probability at least $1 - k\, d^{2N}\delta'$, simultaneously,*

$$\big|(\Gamma_r)_{ij}\big| \;\le\; \epsilon \qquad \textit{for all } r \in [k],\; i, j \in [n]. \tag{25}$$

Proof. Fix r. For $i = j$: $(\Gamma_r)_{ii} = \|S_r u_i\|_2^2 - 1$ and $\|u_i\|_2^2 = 1$, so $|(\Gamma_r)_{ii}| \le \epsilon$ with probability $\ge 1 - \delta'$. For $i \ne j$: orthonormality gives $\|u_i \pm u_j\|_2^2 = 2$, and by the polarization identity together with $\langle u_i, u_j\rangle = 0$,

$$(\Gamma_r)_{ij} \;=\; \langle S_r u_i, S_r u_j\rangle = \tfrac{1}{4}\Big(\big(\|S_r(u_i + u_j)\|_2^2 - 2\big) \;-\; \big(\|S_r(u_i - u_j)\|_2^2 - 2\big)\Big), \tag{26}$$

$$\big|(\Gamma_r)_{ij}\big| \le \tfrac{1}{4}\big(2\epsilon + 2\epsilon\big) \;=\; \epsilon \tag{27}$$

on the intersection of the two embedding events for the fixed vectors $u_i + u_j$ and $u_i - u_j$. Per sketch this is a union over $n + 2\binom{n}{2} = n^2$ fixed vectors; over the k sketches, over $kn^2 = k\, d^{2N}$ events, each of probability at most δ'. □

Remark 5. The absolute value in (25) is essential. The off-diagonal entries of Γ_r are signed; a one-sided bound $(M_r)_{ij} \le \delta_{ij} + \epsilon$ controls neither the lower deviation of E_k nor products containing negative factors, since entry-wise monotone replacement fails for signed products. All estimates below use (25) through the triangle inequality only.

Lemma 6 (Deterministic deviation bound)**.** *Let $\epsilon \ge 0$ and suppose $|(\Gamma_r)_{ij}| \le \epsilon$ for all $r \in [k]$, $i, j \in [n]$. For $\emptyset \ne T = \{r_1 < \cdots < r_j\} \subseteq [k]$ set $q_i := r_{i+1} - r_i$ for $1 \le i < j$ and $q_j := k - r_j + r_1$, so that $q_i \ge 1$ and $\sum_{i=1}^{j} q_i = k$. Then*

$$\big|E_k - \mathrm{Tr}(A^k)\big| \;\le\; \sum_{\emptyset\ne T\subseteq[k]} \epsilon^{|T|} \prod_{i=1}^{|T|} \mathrm{Tr}\big(A^{q_i}\big) \;\le\; \big(\|A\|_2 + \epsilon\, \mathrm{Tr}(A)\big)^k - \|A\|_2^k, \tag{28}$$

and, after dividing by $\mathrm{Tr}(A^k) \ge \|A\|_2^k$,

$$\frac{\big|E_k - \mathrm{Tr}(A^k)\big|}{\mathrm{Tr}(A^k)} \;\le\; k\epsilon \;+\; \sum_{j=2}^{k} \binom{k}{j} (\epsilon\rho)^j \;\le\; k\epsilon + (k\epsilon\rho)^2 \qquad \textit{whenever } k\epsilon\rho \le 1. \tag{29}$$

Proof. Insert $M_r = I + \Gamma_r$ into (19) and expand multilinearly:

$$E_k \;=\; \mathrm{Tr}\Big(\prod_{r=1}^{k}(I+\Gamma_r)\Lambda\Big) \;=\; \sum_{T\subseteq[k]} \mathcal{T}_T, \qquad \mathcal{T}_\emptyset = \mathrm{Tr}(\Lambda^k) = \mathrm{Tr}(A^k). \tag{30}$$

For $T = \{r_1 < \cdots < r_j\}$, collapsing the identity factors merges consecutive powers of Λ, and cyclicity of the trace places exactly q_i factors of Λ between Γ_{r_i} and $\Gamma_{r_{i+1}}$:

$$\mathcal{T}_T \;=\; \mathrm{Tr}\big(\Gamma_{r_1}\Lambda^{q_1}\Gamma_{r_2}\Lambda^{q_2}\cdots\Gamma_{r_j}\Lambda^{q_j}\big) \;=\; \sum_{a_1,\dots,a_j=1}^{n}\;\prod_{i=1}^{j}(\Gamma_{r_i})_{a_i a_{i+1}}\,\lambda_{a_{i+1}}^{q_i}, \qquad a_{j+1} := a_1. \tag{31}$$

By the triangle inequality, $|(\Gamma_{r_i})_{a_i a_{i+1}}| \le \epsilon$, $\lambda_a \ge 0$, and the fact that each summation index a_{i+1} occurs in exactly one eigenvalue factor,

$$|\mathcal{T}_T| \;\le\; \epsilon^j \sum_{a_1,\dots,a_j=1}^{n}\;\prod_{i=1}^{j}\lambda_{a_{i+1}}^{q_i} \;=\; \epsilon^j \prod_{i=1}^{j}\Big(\sum_{a=1}^{n}\lambda_a^{q_i}\Big) \;=\; \epsilon^j \prod_{i=1}^{j}\mathrm{Tr}\big(A^{q_i}\big). \tag{32}$$

Summing over $T \neq \emptyset$ yields the first inequality in (28). Since $\lambda_a^q \le \|A\|_2^{q-1}\lambda_a$,

$$\mathrm{Tr}(A^q) \;\le\; \|A\|_2^{q-1}\,\mathrm{Tr}(A), \qquad \text{hence} \qquad \prod_{i=1}^{j}\mathrm{Tr}(A^{q_i}) \;\le\; \|A\|_2^{k-j}\,\big(\mathrm{Tr}\,A\big)^j, \tag{33}$$

and there are $\binom{k}{j}$ subsets of size j, so the middle term of (28) is at most $\sum_{j=1}^{k}\binom{k}{j}\epsilon^j(\mathrm{Tr}\,A)^j\|A\|_2^{k-j} = (\|A\|_2 + \epsilon\,\mathrm{Tr}\,A)^k - \|A\|_2^k$.

For (29): if $|T| = 1$ then $q_1 = k$ and (32) reads $|\mathcal{T}_T| \le \epsilon\,\mathrm{Tr}(A^k)$; there are k such terms. For $j \ge 2$, by (33) and $\|A\|_2^k \le \mathrm{Tr}(A^k)$,

$$\epsilon^j \prod_{i=1}^{j}\mathrm{Tr}(A^{q_i}) \;\le\; (\epsilon\rho)^j\,\|A\|_2^k \;\le\; (\epsilon\rho)^j\;\mathrm{Tr}(A^k). \tag{34}$$

Finally, for $x := \epsilon\rho$ with $kx \le 1$,

$$\sum_{j=2}^{k}\binom{k}{j}x^j \;=\; (1+x)^k - 1 - kx \;\le\; e^{kx} - 1 - kx \;\le\; (kx)^2, \tag{35}$$

using $e^t \le 1 + t + t^2$ for $t \in [0,1]$. □

Proof of Theorem 3. By Lemma 4 and the choice $\delta' = \delta/(kd^{2N})$, the event (25) holds with probability at least $1-\delta$. On this event apply Lemma 6: the choice (22) gives $k\epsilon \le \hat\epsilon/2$ and $(k\epsilon\rho)^2 \le \hat\epsilon/2 \le 1$ (in particular $k\epsilon\rho \le 1$), hence by (29)

$$\big|E_k - \mathrm{Tr}(A^k)\big| \;\le\; \Big(\frac{\hat\epsilon}{2} + \frac{\hat\epsilon}{2}\Big)\,\mathrm{Tr}(A^k) \;=\; \hat\epsilon\;\mathrm{Tr}(A^k). \tag{36}$$

For (24), substitute (22) into $m = \Theta(N\log(1/\delta')/\epsilon^2)$: $\epsilon^{-2} = \max\{4k^2\hat\epsilon^{-2},\; 2k^2\rho^2\hat\epsilon^{-1}\}$ and $N\log(1/\delta') = N\log(kd^{2N}/\delta) = 2N^2\log d + N\log(k/\delta)$. □

Remark 7 (Comparison; consistency at $k=1$; low rank). Since $\mathrm{Tr}(A^k) \le \|A\|_2^{k-1}\,\mathrm{Tr}(A)$, the prefactor of the bound obtained via the majorization $\mathrm{Tr}(A^p) \le (\mathrm{Tr}\,A)^p$ satisfies

$$\frac{(\mathrm{Tr}\,A)^{2k}}{\big(\mathrm{Tr}(A^k)\big)^2} \;\ge\; \rho^{\,2k-2}, \tag{37}$$

whereas (24) carries $\max\{\hat\epsilon^{-2},\,\rho^2\hat\epsilon^{-1}\}$; e.g. for A a rank-r orthogonal projection the respective prefactors are $r^{2k-2}\hat\epsilon^{-2}$ versus $\max\{\hat\epsilon^{-2}, r^2\hat\epsilon^{-1}\}$. At $k=1$ the expansion (30) contains only the $j=1$ term, (29) degenerates to $|E_1 - \mathrm{Tr}\,A| \le \epsilon\,\mathrm{Tr}(A)$, and Theorem 1 is recovered with no ρ-dependence. If $\mathrm{rank}(A) = r$ is known, every index a with $\lambda_a = 0$ drops out of (31) (each a_i carries a strictly positive power of λ_{a_i}), so Lemma 4 is needed only on the r eigenvectors spanning range(A); replacing d^{2N} by r^2 in δ' replaces $N^2\log d$ by $N\log r$ in (24).

Remark 8 (Tightness within element-wise control; beyond it). Inequality (32) is attained, simultaneously for every T, by $\Gamma_r = \epsilon \mathbf{1}\mathbf{1}^T$: with $J = \mathbf{1}\mathbf{1}^T$,

$$\operatorname{Tr}\left(J\Lambda^{q_1} J\Lambda^{q_2} \cdots J\Lambda^{q_j}\right) = \prod_{i=1}^{j} \mathbf{1}^T \Lambda^{q_i} \mathbf{1} = \prod_{i=1}^{j} \operatorname{Tr}\big(A^{q_i}\big), \tag{38}$$

so the first inequality in (28) holds with equality and Lemma 6 cannot be improved given only (25). The $\rho^2\hat{\epsilon}^{-1}$ term in (24) is likewise unavoidable in this paradigm: for $k = 2$ and the spiked spectrum $\lambda = (1, \mu, \ldots, \mu)$ with $n\mu \to \infty$ and $n\mu^2 \to 0$ (e.g. $\mu = n^{-3/4}$), the $j = 2$ term equals $\epsilon^2(\operatorname{Tr} A)^2 = (1 + o(1))\,\epsilon^2\rho^2 \operatorname{Tr}(A^2)$, forcing $\epsilon \lesssim \sqrt{\hat{\epsilon}}/\rho$. Improving on Theorem 3 therefore requires information about $S^T S$ finer than element-wise control of its Gram action — i.e. moment information. For i.i.d. sketches the exact second moment is

$$\mathbb{E}\big[E_k^2\big] \;=\; \operatorname{Tr}\Big(\big(M\,(A \otimes A)\big)^k\Big), \qquad M := \mathbb{E}\big[S^T S \otimes S^T S\big], \tag{39}$$

so Chebyshev's inequality applied to $\operatorname{Var}(E_k) = \operatorname{Tr}\left((M(A \otimes A))^k\right) - \operatorname{Tr}(A^k)^2$ — the quantity computed by the transfer-matrix analysis of M — yields sample complexities that are instance-optimal at second order, at the price of polynomial (rather than logarithmic) dependence on $1/\delta$.

Remark 9 (Relation to the OSI framework of arXiv:2508.21189). Camaño–Epperly–Meyer–Tropp define an (r, α)-*oblivious subspace injection* (OSI) by isotropy, $\mathbb{E}\|\Omega^* x\|_2^2 = \|x\|_2^2$, together with injectivity: $\sigma_{\min}^2(\Omega^* Q) \geq \alpha$ with probability 19/20 for every fixed orthonormal $Q \in \mathbb{R}^{n \times r}$ — a one-sided *lower* bound with deliberately uncontrolled dilation. The failure mode of E_k is the opposite tail (inflation of the trace through dilation of sketched inner products), so the OSI property alone cannot produce the two-sided guarantee (23). Consistently, their trace-estimation application (§6.2, partition functions) controls error through the estimator's second moment (variance-reduced Girard–Hutchinson with Khatri–Rao probes) combined with OSI-based low-rank deflation — in the present setting, precisely the route (39). One ingredient does transfer: their fourth-moment OSI criterion (§7.1, after Oliveira) requires $\sup_{\|u\|_2=1} \mathbb{E}\|Su\|_2^4 \leq h$, and under the isotropic normalization $\mathbb{E}[S^T S] = I$,

$$\sup_{\|u\|_2=1} \mathbb{E}\,\|Su\|_2^4 \;=\; \sup_{\|u\|_2=1} \;(u \otimes u)^T M\,(u \otimes u) \;\leq\; \|M\|_2 \;=\; R(E^N), \tag{40}$$

so the closed form for the maximal row sum $R(E^N)$ certifies the fourth-moment hypothesis (with $h = R(E^N)$), hence the OSI parameters of the chain sketch — useful for low-rank approximation corollaries, but yielding no improvement to (24).

1.3 Numerical Verification at $k = 3$

Setup. All experiments use the chain sketch with i.i.d. *Rademacher* cores, entries uniform on $\{\pm m^{-1/2}\}$, all bond dimensions equal to m and one output leg of size m, so that $\mathbb{E}[S^T S] = I_{d^N}$ exactly. Rademacher cores realize $\sigma_4 = \sigma_2^2$, the minimum permitted by Jensen's inequality; the sufficiency proof behind Definition 2 is unchanged, since each linearized core is a dense i.i.d. subgaussian matrix and therefore satisfies the strong JL-moment hypothesis entering Theorem 3.1 of Ma–Solomonik with the same scaling (Achlioptas 2003; Kane–Nelson 2014). Gaussian cores reproduce all shapes below with larger constants, e.g. $q_{0.9} = 0.54$ vs. 0.44 at $(N, m, k) = (6, 64, 3)$, $A = I$.

The estimator is evaluated by exact tensor-network contraction; no d^N-sized object is ever formed. With $B_{rs} := S_r S_s^T \in \mathbb{R}^{m \times m}$ computed by transfer steps of cost $O(dm^3)$ per site,

$$A = I_{d^N}: \quad E_3 = \operatorname{Tr}(B_{12} B_{23} B_{31}), \qquad A = \mathbf{1}\mathbf{1}^T: \quad E_k = \prod_{r=1}^{k} \|S_r \mathbf{1}\|_2^2, \tag{41}$$

where $\|S\mathbf{1}\|_2^2$ is a length-N vector–MPS contraction; both test matrices are bond-dimension-1 MPOs with $\operatorname{Tr}(I^k) = d^N$, $\rho(I) = d^N$ (flat extreme) and $\operatorname{Tr}\left((\mathbf{1}\mathbf{1}^T)^k\right) = d^{Nk}$, $\rho(\mathbf{1}\mathbf{1}^T) = 1$ (rank-one extreme). The total cost is $O(kNdm^3)$ per draw. The implementation is validated against a dense computation built from the *same* random cores at $N = 3$ (agreement to machine precision for $k \in \{1, 2, 3\}$ and both test matrices, as well as for the spiked interpolation $\mu I + (1 - \mu)P$ available in the code), and against $\mathbb{E}[E_1] = \operatorname{Tr}(A)$. Code: `verify_m_k3.py`, fixed seeds.

Protocol. $q_{0.9}(m)$ denotes the empirical 0.9-quantile of $|E_k/\operatorname{Tr}(A^k)-1|$ over 96–240 independent draws (matching $\delta = 0.1$), and

$$m^*(N) \;:=\; \min\{m \text{ on the grid } \le 512 :\; q_{0.9}(m) \le \hat{\epsilon}\}, \qquad \hat{\epsilon} = 0.2, \tag{42}$$

with censoring reported when the cap is reached. Quantile noise at these sample sizes is ±10–15%, so m^* is resolved to within one (geometric) grid step.

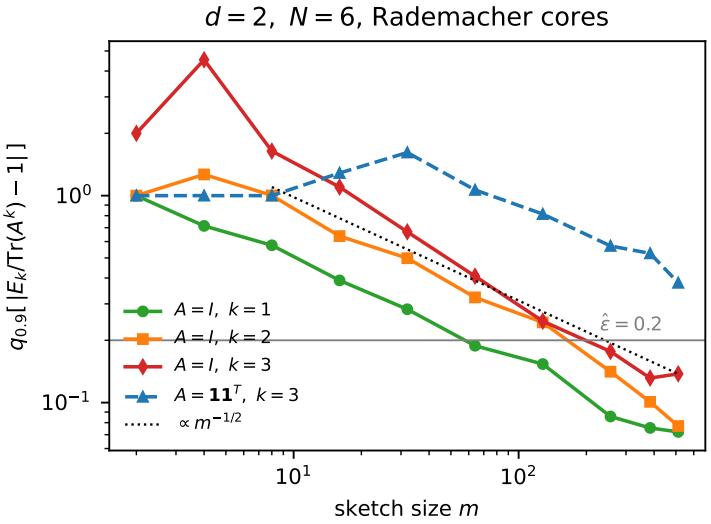


Figure 1: Relative-error quantile vs. sketch size at $d = 2$, $N = 6$. (i) k-dependence on $A = I$: at $m = 64$, $q_{0.9} = 0.19$, 0.32, 0.41 for $k = 1, 2, 3$ — the near-linear growth of the leading $j = 1$ term $k\epsilon$ of Lemma 6; at small m the $k = 3$ curve detaches strongly ($q_{0.9} = 4.5$ at $m = 4$ vs. 0.71 for $k = 1$), the onset of the $j \ge 2$ terms. (ii) All curves decay $\asymp m^{-1/2}$, i.e. $\epsilon = \Theta(\sqrt{N \log(1/\delta')/m})$, the Ma–Solomonik rate. (iii) The rank-one MPO is *uniformly harder* than the identity at equal m: $E_k/\operatorname{Tr}(A^k) = \prod_r \|S_r \mathbf{1}\|_2^2/d^N$ is a pure single-vector embedding problem with no self-averaging. The plateau $q_{0.9} = 1.000$ for $m \le 8$ is exact annihilation: at $d = 2$ the per-site contraction of a ±1 core with $\mathbf{1}_2$ has entries $\xi + \xi' \in \{-2, 0, 2\}$, zero with probability 1/2, so typical draws give $E_k \approx 0$ — the all-ones vector is the adversarial direction for Rademacher entries (cf. the Rademacher Khatri–Rao failure mode in arXiv:2508.21189, §5.3).

N	2	3	4	5	6	7
m^*, $A = I$, $k = 3$	128	256	192	192	256	256
m^*, $A = I$, $k = 1$	48	64	64	96	64	96
m^*, $A = \mathbf{1}\mathbf{1}^T$, $k = 3$	512	> 512	> 512	> 512	> 512	> 512

Sufficiency accounting. At $(\hat{\epsilon}, \delta, k) = (0.2, 0.1, 3)$ the prescription (22)–(24) reads: for $\rho = 1$ (rank one), $\epsilon = \hat{\epsilon}/2k = 1/30$ and $m = \Theta\big(900 \cdot N \log(kd^{2N}/\delta)\big) \approx \Theta(1.1 \cdot 10^4)$ at $N = 2$; for $A = I$ at $N = 6$ ($\rho = 64$), $\epsilon = \sqrt{\hat{\epsilon}/2}/(k\rho) \approx 1.6 \cdot 10^{-3}$ and $m \approx \Theta(2.6 \cdot 10^7)$. Every measured m^* lies *below* the corresponding prescription, by a factor ≈ 20 in the rank-one case and up to $\approx 10^5$ at the flat extreme: the theorem is never violated, and the slack sits exactly where Remark 8 locates it. The empirically binding regime is branch 1 (per-vector embedding accuracy, with the sketch's true constants): the rank-one instances ride it and exhaust the grid. The $\rho^2\hat{\epsilon}^{-1}$ branch is the price of *adversarial* Gram perturbations consistent with element-wise control; the typical randomness of the sketch does not realize it at $A = I$, because the trace self-averages over the d^N eigendirections — the flat $m^*(N)$ in Fig. 2 quantifies this within $N \le 7$. Consequently, the gap between m^* and (24) for flat spectra is not closable by a sharper union bound; it is the second moment (39) that prices typical draws. What the numerics cannot certify are the Θ-constants inherited from Ma–Solomonik and the behaviour beyond $N = 7$; no claim is made there.

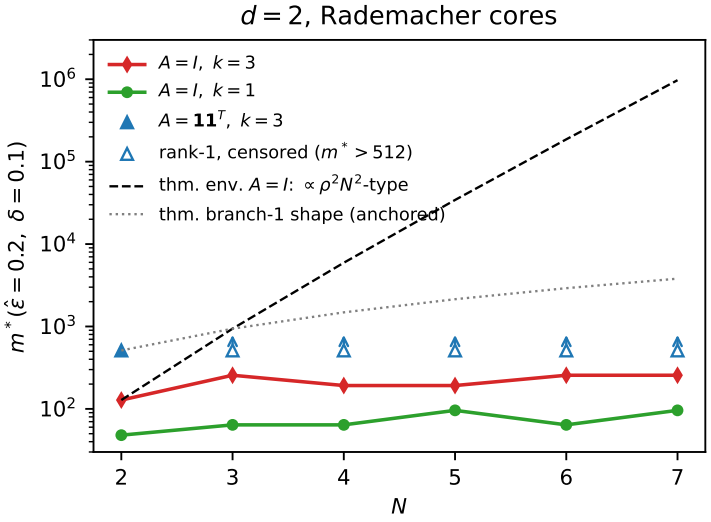


Figure 2: Required sketch size $m^*(\hat{\epsilon} = 0.2, \delta = 0.1)$ vs. N at $d = 2$, $k = 3$ (and $k = 1$ control). Measured m^* for $A = I$ is flat in N over the accessible range; the rank-one case is censored (> 512) for $N \geq 3$, consistent with the branch-1 shape $\propto N \log(kd^{2N}/\delta)$ anchored at $N = 2$ (dotted). The theorem envelope for $A = I$ (dashed, $\propto \rho^2 N^2$-type, anchored from above) majorizes every measurement, as it must.